§5b — Stratification is Forced by an Impossibility (draft for integration)

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Stratification is forced by an impossibility, not chosen as a refinement

The impossibility. An anytime-valid e-process E_t certifies $\mathbb{E}[E_\tau] \le 1$ at every stopping time τ. That guarantee is *marginal*—averaged over the distribution of test points the monitor happens to see. It does not, and cannot distribution-free, certify the conditional statement $\mathbb{E}[E_\tau \mid \text{difficulty} = d] \le 1$ for every difficulty level d. This is the conditional-coverage impossibility of Barber, Candès, Ramdas, and Tibshirani (arXiv:1903.04684; primary read pending): with no assumptions on the data-generating distribution and in finite samples, marginal and conditional validity cannot be had together. The gap is structural, not an artifact of a betting rule.

Why it bites the co-failure monitor specifically. One is tempted to route around difficulty analytically. The natural candidate is the detection drift

$$\mu \;=\; \mathrm{KL}(P_{\text{joint}} \,\|\, P_{\text{marg-product}}),$$

the Kullback–Leibler divergence between the observed joint failure law and the product of its marginals. Because $\mu = 0$ iff the two judges fail independently, μ is a pure dependence functional, and it is tempting to call it "difficulty-orthogonal by construction"—whatever the base failure rate, μ sees only the coupling. This is false, and the way it is false is instructive. For Bernoulli failure indicators, mutual information is not a copula invariant: fix the odds ratio (the coupling) and sweep the marginal failure rate from 0.05 to 0.5, and μ moves by an order of magnitude. Difficulty re-enters the e-process through two channels that do not cancel—the variance of each log-likelihood increment scales with the marginal $p(1-p)$, and estimation error in the nuisance marginals propagates into the realized increments. The only sense in which μ "conditions out" difficulty is circular: define "same coupling" to mean "same μ", which defines the problem away rather than solving it.

Consequence. So the drift law, latency $\propto 1/\mu$, survives as the quantitative heart of the monitor—but not as a difficulty-robust one. Its non-robustness *is* the conditional-coverage gap: no distribution-free functional of the stream can deliver per-difficulty validity for free. Stratification is what the impossibility leaves open—the one escape it does not close. We stratify not to refine the estimator but because the theorem forbids the alternative.

The positive construction. Partition the difficulty axis into K strata and run a separate betting e-variable within each. On items in stratum k,

$$e \;=\; 1 + \lambda\big(U - V - \delta_k\big),$$

where $U = \mathbf{1}\{\text{both judges fail on item } i\}$ and $V = W_i^s W_j^t$ pairs the two judges across *different* items $s \neq t$, so the marginals multiply exactly, $\mathbb{E}[V] = ab$, with no fitted nuisance. The slack $\delta_k = 2\varepsilon$ absorbs the worst-case two-sided excursion of the stratum's base rate, and the betting parameter is confined to $\lambda \in [0, 1/(1+2\varepsilon)]$ so the e-variable stays non-negative. The product over strata (or a mixture across k) is again a valid e-process by Ville. Conditional coverage is recovered by construction—one certificate per stratum—at the $O(K)$ cost the impossibility theorem says such coverage must carry.

The practical difference. On a synthetic panel where difficulty and coupling are confounded, the three methods behave as follows:

Method	Description	Type-I error
A	Stratified cross-item monitor	0.00
B	Unstratified cross-item monitor	0.23–0.44
C	Naïve plug-in (fitted marginals)	0.59–0.70

Stratification is not tightening a bound; it is the difference between a valid test and an invalid one. (Synthetic; a proof of necessity, not an empirical effect size.)

A second conditioning variable, and a distinction we keep. Difficulty is not the only axis on which a panel's ballots carry structure. Shu's pivotality analysis (arXiv:2608.06940) conditions on the vote margin $m_i = |2s_i - k|$: nearly all of a panel's recoverable accuracy sits on the $\approx 16\%$ of queries where a single flip would change the verdict, and, unlike difficulty, margin needs no labels to compute. It is tempting to treat difficulty and margin as interchangeable stratification axes governed by one impossibility. They are governed by the same impossibility—neither can be conditioned out for free—but they are not interchangeable, and the difference is where our construction lives.

Difficulty is *ancillary* to the tested dependence: conditioning on it leaves the two judges' ballots conditionally independent, so $\mathbb{E}[V] = ab$ is preserved stratum by stratum. Margin is *not* ancillary; it is near-sufficient for the co-failure alternative, because co-failure *is* excess agreement and margin is a function of the agreement count. Conditioning on the ballot sum $s_i = c$ pins each judge's marginal to c/k and induces a negative within-item covariance,

$$\mathrm{Cov}(W_i, W_j \mid s_i = c) \;=\; -\frac{(c/k)(1 - c/k)}{k-1} \;<\; 0,$$

which breaks the exact factorization $\mathbb{E}[V] = ab$ that the whole margins-free construction rests on—and drains the test's power precisely at unanimity, where co-failure is strongest. We therefore keep margin as a *validity axis*—a label-free check that Type-I control does not depend on where on the margin scale the panel sits—and not as a power-bearing stratification. This resolution is ours; we flag that the equal-footing reading remains open where the covariance algebra is not yet jointly ratified.

Alternatives under investigation (brief). Two alternatives to explicit stratification are under investigation and not yet adopted: shape-constrained e-processes (Wang–Ramdas, arXiv:2604.20612), which would monitor whether the failure-rate curve keeps its monotone/unimodal shape in difficulty rather than stratifying it out; and DKW-corrected conditional test martingales (arXiv:2602.13848). Both are flagged pending primary reading and are mentioned only for completeness.